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OPTICAL MUSIC RECOGNITION FOR JAZZ LEAD SHEETS: A CRNN-CTC APPROACH FOR MONOPHONIC LINES

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GEP Deliverable 3: Budget and Sustainability

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1 Project Budget

This section presents the full economic budget for the project, following the structure recommended in the GEP economic management module. The budget is broken down into: (1) human resources costs, itemised per Gantt activity; (2) general costs (hardware, software, and overheads); (3) a contingency reserve; and (4) specific incidentals linked to risks identified in Deliverables 1 and 2.

1.1 Human Resources Costs

The project is executed by a single developer who assumes all professional roles. To estimate labour costs realistically, each task is assigned to the role that best describes the work performed, with an hourly rate derived from average gross salaries in Spain for the IT sector. Rates are grounded in the 2023 INE *Encuesta Anual de Estructura Salarial* (EAES) [1], the official Spanish salary survey published by the Instituto Nacional de Estadística. The national mean gross salary was €28,049.94 in 2023; workers in the ICT sector (Section J of NACE rev. 2) earn substantially above the national average, making the €30,000–38,000 gross range assumed here consistent with EAES data for junior-to-mid level ICT roles. These figures are adjusted by a ×1.35 multiplier to convert gross salary to total employer cost, which accounts for employer social security contributions [2], as prescribed in the GEP economic management module [3].

The four roles and their fully loaded hourly rates are:

Table 1: Role definitions and hourly rates (gross salary × 1.35 social security [2]).

Role	Gross salary (€/year)	Loaded (€/year)	€/h[1]
Project Manager	38 000	51 300	29.15
Research Analyst	33 000	44 550	25.31
ML Engineer	36 000	48 600	27.61
Technical Writer	30 000	40 500	23.01

Table 2 shows the aggregated human resources cost per work package; the per-task breakdown is consistent with the Gantt chart in Deliverable 2.

Table 2: Human resources budget by work package.

WP	Work Package	Hours	Cost (€)
WP0	Project Management (planning, meetings, thesis writing)	130	3 482.50
WP1	Sprint 1 – Literature Review & Baseline	78	2 066.20
WP2	Sprint 2 – Dataset Construction	73	1 992.55
WP3	Sprint 3 – CRNN-CTC Development	75	2 075.37
WP4	Sprint 4 – Iterative Refinement	68	1 859.10
WP5	Sprint 5 – Final Evaluation & Defence	85	2 139.85
	TOTAL HUMAN RESOURCES	**509**	**13 615.57**

1.2 General Costs

General costs cover hardware, software, and infrastructure expenses that are shared across all project activities and cannot be attributed to a single task.

[1] Based on 1 760 effective working hours per year (220 days × 8 h/day).

1.2.1 Hardware

Table 3: Hardware costs with amortisation.

Item	Price (€)	Life (yr)	Use (mo)	Amort. (€)
Desktop PC (with RTX 3060 12 GB)	1 200	5	5	100.00
Monitor (external)	250	6	5	17.36
Peripherals (keyboard, mouse)	80	4	5	8.33
Total hardware amortisation				**125.69**

Amortisation is calculated linearly: $\text{cost} = \text{price} \times \frac{\text{project months}}{\text{useful life} \times 12}$.

1.2.2 Software

All software used in this project is free/open-source or available through university licences at no additional cost:

Table 4: Software costs.

Software	Licence	Cost (€)
Python 3.14, PyTorch, OpenCV, music21	Open-source	0.00
LilyPond + LilyJAZZ	Open-source (GPL)	0.00
LaTeX (TeX Live)	Open-source	0.00
Git / GitHub	Free (education)	0.00
Zotero	Open-source	0.00
Linux (Fedora)	Open-source	0.00
Total software		**0.00**

1.2.3 Other Overheads

Table 5: Infrastructure and overhead costs (5-month project duration).

Item	Monthly (€)	Justification	Total (€)
Electricity	15.00	Desktop PC + GPU ≈ 300 W × 5 h/day	75.00
Internet	10.00	Proportional share of home broadband	50.00
Workspace (home)	0.00	No additional rent (home office)	0.00
Cloud GPU (reserve)	–	Only if R4 materialises (see §1.4)	0.00
Total overheads			**125.00**

1.3 Contingency

A contingency reserve of **15%** is applied to the sum of human resources and general costs. This percentage is within the 10–20% range recommended for IT projects and reflects the moderate level of detail in this budget (task-level granularity, but single-person execution with inherent estimation uncertainty).

$$\text{Contingency} = 0.15 \times (13\,615.57 + 250.69) = 0.15 \times 13\,866.26 = \textbf{€2 079.94}$$

1.4 Incidentals

Incidental costs are derived from the risks identified in Deliverables 1 and 2 (Table 4 of Deliverable 2). Each incidental is costed at its full remediation cost and then weighted by the probability of occurrence.

Table 6: Incidentals budget linked to project risks.

Risk	**Description**	**Remediation action**	**Full cost (€)**	**Prob.**	**Weighted (€)**
R1	Limited ground-truth data	Fall back to PrIMuS typeset images; 10 h extra augmentation work.	276.10	25%	69.03
R3	Model underfitting / CTC misalignment	Simplify vocabulary and retrain; 10 h extra ML Engineer time from T5.3 buffer.	276.10	20%	55.22
R4	Computational constraints	Rent cloud GPU (e.g., Lambda or Vast.ai) for 40 h at ∼€1/h + 5 h setup.	178.05	15%	26.71
R5	Scope creep	Reduce T5.3 to feasibility discussion (2–3 h writing); no extra hardware cost, only re-planned hours already budgeted.	0.00	30%	0.00
Total incidentals					**150.96**

1.5 Total Budget

Table 7: Total project budget.

Budget item	**Cost (€)**
Human resources (Table 2)	13 615.57
General costs (hardware, software, overheads)	250.69
Contingency – 15% (§1.3)	2 079.94
Incidentals (Table 6)	150.96
TOTAL PROJECT BUDGET	**16 097.16**

The total estimated project cost is **16 097.16 €**. Human resources represent 84.6% of the total, which is expected for a research-intensive software project with minimal hardware requirements. The contingency and incidentals together account for 2 230.90 € (13.9%), providing a reasonable safety margin.

2 Sustainability Report – Initial Milestone

This section addresses the initial-milestone sustainability questions (row "I" of the sustainability matrix defined in the GEP sustainability guide [4]; see also the economic management module [3]). The analysis covers the three mandatory dimensions: economic, environmental, and social.

2.1 Self-Assessment of Sustainability Competency

After completing the sustainability self-assessment survey, my main strengths are in economic dimension awareness and understanding energy consumption in computational workloads. Areas for growth include full lifecycle impact analysis (supply chain, e-waste) and reasoning about broader societal implications beyond direct user benefits – aspects this project has prompted me to consider more carefully.

2.2 Economic Dimension

PPP – Project cost estimation. The total estimated cost of the project is 16 097.16 € (see §1.5). Human resources dominate the budget (84.4%), reflecting the research and development nature of the work. Material costs are low thanks to the exclusive use of open-source software and existing consumer hardware. The budget includes a 15% contingency reserve and risk-weighted incidentals, providing a realistic safety margin.

Useful life – Comparison with the state of the art. Currently, musicians and archivists who need to digitise jazz lead sheets must either transcribe scores manually (slow and expensive, typically €15–30 per page for a professional copyist) or use commercial OMR tools such as Audiveris, SmartScore, or PhotoScore, which range from free (with limited accuracy) to €100–250 for a licence. However, none of these tools are specifically tuned for the jazz lead-sheet idiom (handwritten-style fonts, chord symbols, specific clefs), resulting in high error rates that still require significant manual correction.

This project aims to produce an open-source, specialised model that, once trained, can be run at negligible marginal cost: inference on a single score image takes under one second on consumer hardware. If the model achieves a low SER, it would substantially reduce the cost of digitising lead-sheet collections compared to both manual transcription and generic commercial tools. The model weights, code, and training pipeline will be released publicly, eliminating licence costs for downstream users.

2.3 Environmental Dimension

PPP – Environmental impact estimation. The main environmental impact of the project is electrical energy consumption for computation and personal work. We estimate:

- **Development workstation:** The desktop PC (excluding GPU) and monitor consume approximately 80 W during regular development (coding, writing, literature review). Over 509 project hours, this amounts to $509 \times 0.08\,\mathrm{kW} = 40.7\,\mathrm{kWh}$.
- **GPU training:** The NVIDIA RTX 3060 (integrated in the same desktop) draws $\sim$170 W under training load. Estimated GPU-active time across all training tasks (T3.3, T4.3, T4.4, T5.1, T5.3): $\sim$95 h, yielding $95 \times 0.17\,\mathrm{kW} = 16.2\,\mathrm{kWh}$.
- **Dataset generation:** LilyPond rendering of $\sim$43 000 images is CPU-bound; estimated at $\sim$10 h of sustained CPU ($\sim$65 W), contributing 0.65 kWh.

- **Total:** $\sim$57.5 kWh over five months, equivalent to approximately 13.2 kg CO_2 at the 2023 Spanish grid emission factor of 0.23 kg CO_2/kWh [5], [6].

To put this in context, 57.5 kWh is roughly equivalent to running a household refrigerator for 4–5 days or driving a petrol car for approximately 55 km. The environmental footprint is modest.

PPP – Minimisation measures. Several measures are taken to reduce the environmental impact:

- **Efficient hardware:** The RTX 3060 was chosen for its strong performance-per-watt ratio, and the ResNet18 backbone is deliberately lightweight (vs. larger architectures like ResNet50 or Vision Transformers).
- **Mixed-precision training:** PyTorch AMP (Automatic Mixed Precision) reduces GPU memory usage and speeds up training, lowering total energy consumption per training run.
- **Early stopping:** Training halts automatically when validation loss plateaus, avoiding unnecessary computation.
- **Re-use of resources:** No new hardware was purchased for this project; all equipment was already owned and in use.

Useful life – Comparison with the state of the art. Manual transcription has negligible direct energy cost but consumes significant human time. Commercial OMR tools run on standard PCs with similar energy profiles. Cloud-based OMR services (e.g., Google Cloud Vision for document OCR) involve data-centre energy consumption that is typically higher per query than local inference. This project's model runs locally on consumer hardware, avoiding data-centre overhead. Furthermore, by releasing a trained model, the environmental cost of training is incurred *once* and amortised across all users, rather than each institution training its own model.

2.4 Social Dimension

PPP – Personal impact. This project has deepened my expertise in deep learning, computer vision, and music information retrieval, and strengthened my ability to plan and execute a medium-scale research project independently. As a musician, it has also sharpened my awareness of accessibility challenges faced by visually impaired musicians and the importance of machine-readable music formats for assistive technologies.

Useful life – Comparison with the state of the art and real need. Jazz lead sheets from The Real Book are among the most widely used musical documents in jazz education worldwide, yet remain largely in paper or scanned-image form. Existing OMR tools perform poorly on the jazz lead-sheet idiom (handwritten-style fonts, chord symbols, informal layout), forcing musicians and archivists to fall back on manual transcription. The direct beneficiaries are musicians (transposition, digital annotation), educators (interactive materials), archivists (searchable collections), and MIR researchers (symbolic data for computational musicology). No group is adversely affected: the system processes user-owned scores, collects no personal data, and is released open-source.

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